Programming with Generative AI

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Lecture 1:

- **What** do programmers do?
- **Why** is Gen AI so good at doing those things?



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What do programmers do?

Q: What do swimmers do?
A: They wave their arms and legs 

Programmers write code

Imagenet classification with deep convolutional neural networks
A Krizhevsky, I Sutskever… - Advances in neural …, 2012 - proceedings.neurips.cc
We trained a large, deep convolutional neural network to classify the 1.3 million high-resolution images in the LSVRC-2010 ImageNet training set into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 39.7\% and 18.9\% which is considerably better than the previous state-of-the-art results. The neural network, which has 60 million parameters and 500,000 neurons, consists of five convolutional layers, some of which are followed by max-pooling layers, and two globally connected layers with a final …
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ertain types)

anet beyond Neptune

- *Problem*

Data-ce

PRIMES is in **P**
M Agrawal, N Kayal, N Saxena - Annals of mathematics, 2004 - JSTOR
… We give a deterministic, O'(log15/2 n) time algorithm for testing if a number is **prime**. Heuristically, our algorithm does better: under a widely believed conjecture on the density of …
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1. Suppose the given integer is *n*.
2. If *n* is less than 2, say "No" and stop.
3. For each integer *x* from 2 to *n* – 1:
   If *x* is a factor of *n*, say "No" and stop.
4. Say "Yes" and stop.

Test Your Understanding

- What do the words “algorithm” and “algebra” have in common? (*Hint*: al-Khwārizmī)
- **Algorithm**: A **finite** sequence of **basic** instructions

Algorithm 1:
Check if a given number is prime.

1. Suppose the given integer is n.
2. If n is less than 2, say “No” and stop.
3. For each integer x from 2 to $n - 1$:
   If x is a factor of n, say “No” and stop.
4. Say “Yes” and stop.

Algorithm 2:
Find the smallest prime number larger than a given number.

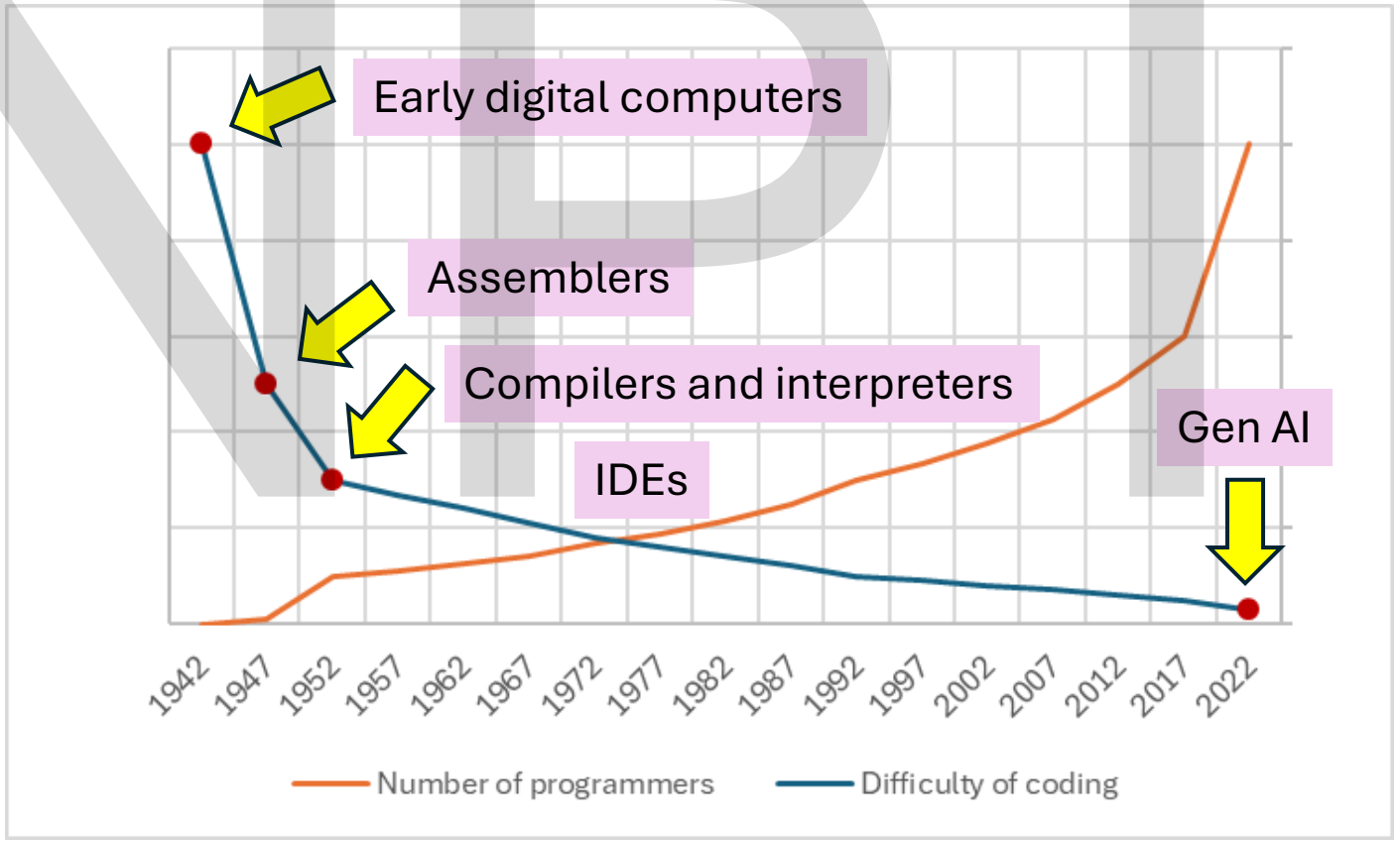
1. Suppose the given integer is n.
2. For each integer x from n+1 to ∞:
   If x is a prime number, print x and stop.

x is a factor of n?



How hard is it to write code?

- 1940's to present (*indicative*)



Natural language

Comment

```
# The answer is 42
```

Hallucination?

```
answe = 42
```

```
0100 0010 1110 1010
```

```
movl   $42, %eax
```

```
0100 0010 1110 1010
```

```
0100 0010 1110 1010
```

We must **critique** and **test** AI-generated code

Evaluating large language models trained on code

M Chen, J Tworek, H Jun, Q Yuan, HPDO Pinto… - arXiv preprint arXiv …, 2021 - arxiv.org

… for **evaluation** so as to not overwhelm them. Inspired by similar work in **language modeling**, … with the highest mean token log probability outperforms **evaluating** a random sample, while …

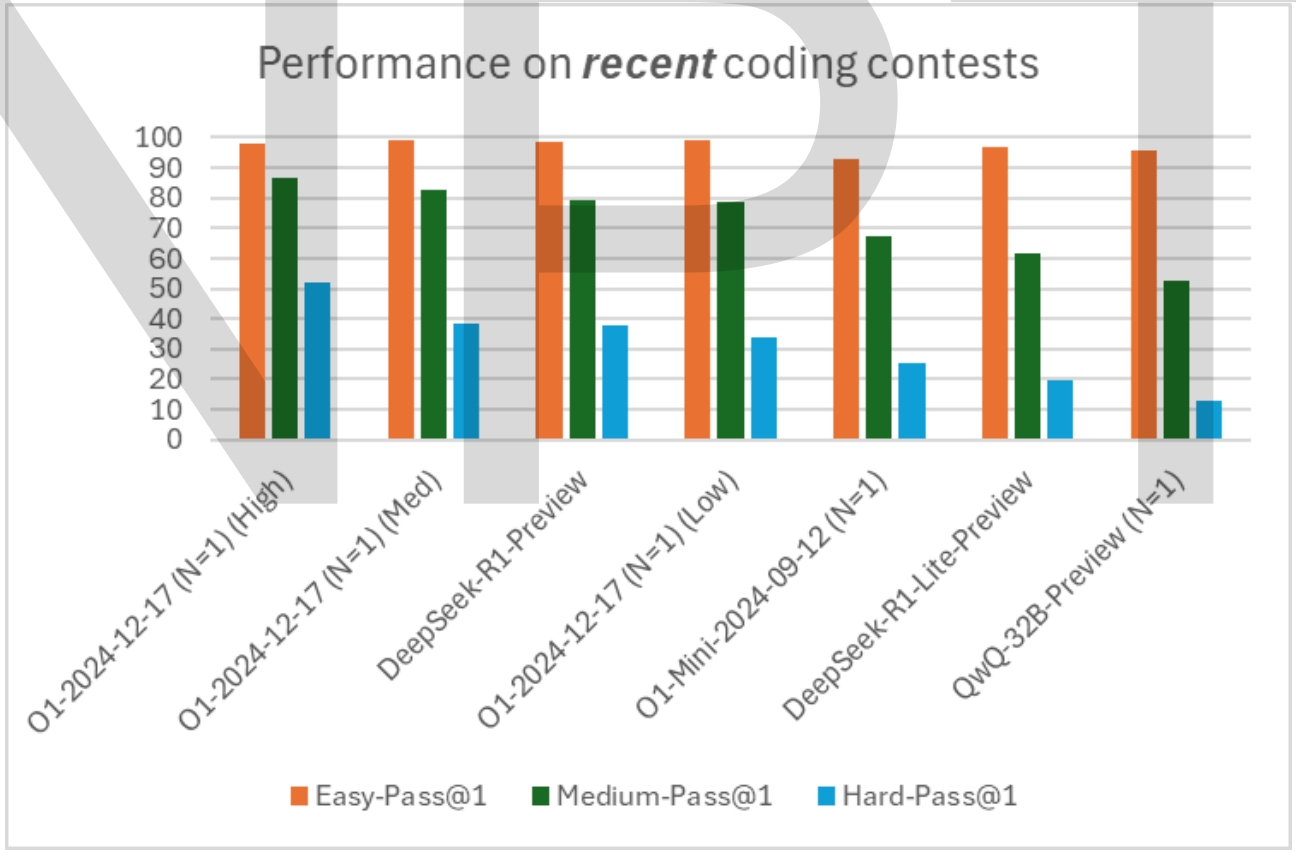
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One of the key risks associated with using code generation models in practice is over-reliance on generated outputs. Due to the limitations described above as well as alignment issues described below, Codex may suggest solutions that superficially appear correct but do not actually perform the task the user intended. This could particularly affect novice programmers, and could have significant safety implications

4. Over-reliance on Gen AI may lead to shallow learning.

How effective is Generative AI at coding?

Source: LiveCodeBench Leaderboard



Sample "Easy" problem

Slavic is preparing a present for a friend's birthday. He has an **array *a* of *n* digits** and the present will be the **product of all these digits.** Because Slavic is a good kid who wants to make the biggest product possible, he wants to **add 1 to exactly one of his digits.**

What is the maximum product Slavic can make?

Input:

```
[4, 3, 2, 3, 4]
```

Output:

```
432
```

$= 4 \times 3 \times \mathbf{3} \times 3 \times 4$

Test Your Understanding

- Critique the following "algorithm" to solve the sample "Easy" problem:

- Input: Digit sequence $a_1, a_2, \ldots, a_n$ (e.g., 4, 3, 2)
1. Find the unique digits (e.g., 2, 3, 4)
2. Sort them in descending order (e.g., 4, 3, 2)
3. "Glue" them as an integer x (e.g., 432)
- Output: x



concatenation

Input

Output

We **refute** the **buggy** algorithm by providing an **Input** on which the algorithm's **Output** is **not** the **Expected Output**

Expected Output

$5 \times 3 \times 2 = 30$

$4 \times \mathbf{4} \times 2 = 32$

$4 \times 3 \times \mathbf{3} = 36$

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Why is Generative AI so effective?

- Large volumes of code that is:
  - Open-source
  - Well-structured
  - Well-documented

Source: The Python Programming Language

```
311     static inline int
312     valid_index(Py_ssize_t i, Py_ssize_t limit)
313     {
314         /* The cast to size_t lets us use just a single comparison
315            to check whether i is in the range: 0 <= i < limit.
316
317            See:  Section 14.2 "Bounds Checking" in the Agner Fog
318            optimization manual found at:
319            https://www.agner.org/optimize/optimizing_cpp.pdf
320         */
321         return (size_t) i < (size_t) limit;
322     }
```

- Used to train Gen AI models
  - Open-source piracy? (e.g., GitHub Copilot litigation)
  - "Free" vs. business versions (e.g., training GitHub Copilot on user's code)

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A (current) weakness of Generative AI

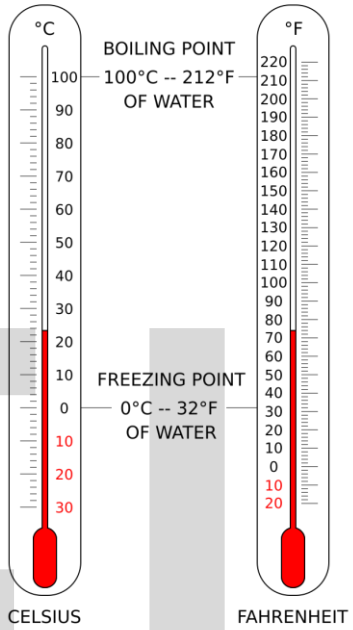
- If the problem is unclear, it makes **assumptions**:

We must learn to ask **clarifying questions**

Write a Python function to convert temperature in °C to °F

Wait! Before I provide an answer, I need to clarify a few things:

1. What is the **datatype** of the temperature in °C?
2. What should I do if you give me a temperature that is below absolute zero (–273.15 °C)?



Source: Wikipedia

Test Your Understanding

- Consider the following problem: *Find the smallest number in a given list*.
- Which of these are **good** clarifying questions?

1. What if the list is empty?
2. What if the list has no numbers?
3. What if all the numbers in the list are unique?
4. What if the smallest number appears more than once in the list?

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Summary

- Programmers write code to solve problems with two types of solutions: **algorithm**-centric & **data**-centric
- Gen AI continues a trend towards **democratizing** programming (easier for more and more people)
  - It can solve non-trivial problems (but legality is a concern)
- To be effective at programming with Gen AI, we must learn to deal with:
  - AI **hallucinations**: by **critiquing** and **testing** code AI-generated code
  - AI **assumptions**: by asking **good** **clarifying questions**

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